Sentinel - Business Plan

Market opportunity, business model, competition, and milestones

Prepared for incubation evaluation | 2026 | Companion: Executive Summary, White Paper

1. Market Opportunity

The mental-health technology market is growing at double-digit rates globally, driven by clinician shortage, rising demand for continuous care, and post-pandemic acceptance of digital mental health. Sentinel targets the underserved middle: institutions that need real-time, clinically integrated monitoring but cannot accept per-seat SaaS costs, cloud-data residency risks, or black-box AI decisions.

Segment	Who they are	What Sentinel offers
Counselling centres & clinics	Private practices, hospital psychiatry depts	Between-session monitoring, crisis safety net, triage dashboards
Universities & campuses	Student mental health offices	Low-cost deployment, offline privacy, trusted-contact loop
Low-resource / rural settings	Clinics with limited IT staff	Zero-infrastructure cost, commodity hardware, PWA access
Research institutions	Psychophysiology / HCI labs	Open, auditable pipeline; ring SDK; event store for studies

2. Business Model

- Primary: on-premises software license + support for clinics and institutions (per-site annual license, unlimited users/seats).
- Hardware bundle: optional smart-ring kit (BLE gateway + rings) for programs that want turnkey physical deployment; margin on hardware.
- Research tier: free academic license in exchange for published validation data (cohort studies).
- Cost structure is exceptional: software stack, storage, and local AI are zero-cost; the platform has no recurring cloud bill, enabling thin margins at low prices.

Revenue levers are the software license, the hardware bundle, and later a managed hosted tier for customers who prefer SaaS over on-prem. Offline-first architecture means even the hosted tier uses the customer's local inference, keeping marginal compute cost near zero.

3. Competition

Competitor	Model	Gap Sentinel fills
Wellness apps (generic)	B2C subscriptions	No clinician integration, no real-time escalation
Telehealth platforms	Per-session B2B	Appointment-only; nothing between sessions
Mental-health chatbots	B2C AI chat	No physiology, no clinical workspace, black-box AI
Wearable vendor dashboards	B2C analytics	Patient-only data; no psychologist or contact loop
Crisis helplines	Public service	Standalone; no continuity of care

No competitor combines continuous physiology, emotion-labelled language, a deterministic crisis escalation loop, an offline AI path, and a tri-directional stakeholder loop in a single auditable system.

4. Go-to-Market and Milestones

- Phase 1 (now): pilot with 1-2 counselling centres / a university counselling office; hardware M0 complete.

- Phase 2 (next): clinical validation study with a partner institution; live BLE/vendor ring ingestion (M1).
- Phase 3: campus-scale deployment, published validation, research-tier academic adoption.
- Phase 4: managed hosted tier + FHIR-adjacent export for regulated clinical use.

5. Financial Summary

Item	Assumption	Value
Operating cost	Open-source stack + local AI + SQLite/PostgreSQL	$0/month recurring
Revenue stream	Per-site annual license (pilots at intro pricing)	TBD at pilot
Hardware bundle	OEM ring + BLE gateway per program	Margin on cost
Funding ask	Pilot infrastructure + partner access + regulatory mentorship	Incubation program

6. Risks and Mitigations

- Clinical acceptance: mitigated by explainable, deterministic safety paths and AI-as-assistant posture.
- Regulatory path: mitigated by on-prem data residency, audit log, signed links, and encryption defaults.
- Hardware dependency: mitigated by vendor-agnostic RingSource SDK and a full deterministic simulator for software-only pilots.

Full technical detail, security posture, and measured performance are documented in the White Paper and Validation & Timing Report.